Ravinder Varkali

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SENIOR STAFF SOFTWARE ENGINEER | DISTRIBUTED SYSTEMS | PLATFORM ENGINEERING | CLOUD

Senior Staff Software Engineer specializing in distributed systems, platform engineering, cloud-native architecture, observability, reliability, and security. Proven track record of architecting scalable microservices and event-driven systems using Go, Java/Spring Boot, Kubernetes, Terraform, PostgreSQL, Redis, and Kafka across AWS, GCP, and Azure. Hands-on technical leader experienced in designing highly available platforms, driving architecture and engineering standards, and leading complex initiatives across engineering organizations.

CORE COMPETENCIES

Backend & Distributed Systems: Java, Spring Boot, Go, Microservices, Event-Driven Architecture, Distributed Workflows, gRPC, REST APIs, High Availability, Scalability, Fault Tolerance, Design Patterns
Cloud & Platform: Kubernetes, Docker, Terraform, GCP, AWS, CI/CD, Infrastructure as Code
Observability & Reliability: OpenTelemetry, Distributed Tracing, Metrics, Logging, Grafana, Elasticsearch
Data & Messaging: PostgreSQL, Redis, Kafka, Amazon Kinesis, DynamoDB
Security: Zero Trust, OAuth2/OIDC, JWT, RBAC, ABAC, ReBAC, mTLS, Authentication & Authorization, Least Privilege
Languages: Java, Go, Python, JavaScript/TypeScript
Technical Leadership: Platform Architecture, System Design, Architecture Reviews, Engineering Standards, Cross-functional Leadership, Mentoring
Applied AI: LLM Integration, Tool/Function Calling, Model Context Protocol (MCP)

PROFESSIONAL EXPERIENCE

ZSCALER, INC. — San Jose, CA

Senior Staff Software Engineer | Mar 2024 – Present

- Architected and led implementation of ZObserver, a distributed observability and troubleshooting platform supporting services across Zscaler Cloud, reducing troubleshooting time and operational costs by 30%.
- Architected distributed orchestration across controllers and large-scale agent fleets using asynchronous messaging, parallel/sequential workflows, concurrency controls, rate limiting, retries, and failure isolation.
- Validated platform scalability through load testing at 500K simulated cloud servers, exercising distributed orchestration, messaging, and agent communication under large-scale workloads.
- Improved end-to-end diagnostic execution latency from ~6 minutes to ~40 seconds by optimizing scheduling, timeout handling, batching, and Redis communication patterns.
- Reduced messaging and connection pressure through agent-side batching, Redis pipelining, transient-data TTL strategies, and separation of persistent execution data from messaging infrastructure.
- Extended the platform with RBAC-secure, AI-assisted engineering workflows, exposing controlled service-discovery and diagnostic capabilities through reusable tool interfaces.

VMWARE, INC. — Palo Alto, CA

Staff Software Engineer | Jul 2020 – Nov 2023

- Architected and developed distributed, event-driven backend services for VMware Cloud Partner Navigator, including audit logging, partner-order telemetry, and cloud-service integrations supporting enterprise-scale SaaS operations and partner lifecycle management.
- Improved customer-support portal performance by 40% through parallel and asynchronous processing.
- Integrated Dell/VMware Cloud service workflows, improving order and subscription visibility for partners.

NETSKOPE, INC. — Santa Clara, CA
Staff Software Engineer | Jan 2020 – Jul 2020

- Built internal engineering tools and infrastructure for development, debugging, scaling, and monitoring of the Netskope SaaS platform.
- Integrated proactive alerting and observability using Prism, SignalFx, and Grafana.

ASURION, LLC. — San Mateo, CA
Principal Software Engineer | Mar 2015 – Dec 2019

- Architected multi-tenant, customer-facing mobile backup and restore platforms using horizontally scalable distributed microservices deployed on Kubernetes.
- Designed asynchronous, event-driven processing using AWS Kinesis and Redis, supporting horizontally scalable workloads and decoupled service execution.
- Designed resilient long-running backup/restore workflows with retry handling, partial-failure recovery, idempotent processing, and scalable service coordination.
- Developed Java/Spring Boot REST APIs for search, albums, tags, authentication, and RBAC-based authorization; built facial-recognition image search using Amazon Rekognition.

WIPRO TECHNOLOGIES — Cupertino, CA
Technical Lead | Dec 2005 – Feb 2015

Key engineering engagements: Apple Inc. and global financial-services clients

- Designed a data-browser reporting platform for Apple's Global Business Intelligence organization and implemented metadata and search APIs backed by SAP HANA.
- Led delivery of enterprise analytics, reporting, metadata search, and Java/J2EE applications for global financial-services and technology clients.

IBM SOFTWARE LABS. — Bangalore, India
Software Engineer | Dec 2003 – Dec 2005

- Developed enterprise portlets and SAP and email integrations for IBM WebSphere Portal Server.
- Delivered new capabilities for IBM WebSphere Application Server.

SELECTED ENGINEERING PROJECTS

- **Enterprise IAM & Zero Trust:** Spring Boot, OAuth2/OIDC, OpenFGA, OPA, mTLS; identity and policy-based authorization.
- **AgentTrust — AI Agent Security**: MCP, least-privilege tool authorization, security controls.
- **gRPC Microservices**: Go, gRPC, PostgreSQL, OpenTelemetry; distributed service architecture.

PATENT APPLICATION

Co-inventor, "Cloud Observability Framework for Providing Troubleshooting and Monitoring for Services" — U.S. Patent Application No. 19/069,767, filed 2025, covering distributed diagnostics, cloud observability, and troubleshooting workflows.

EDUCATION

M.Tech. in Computer Science and Engineering — Indian Institute of Technology Madras
B.Tech. in Civil Engineering — Osmania University, Hyderabad

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate
- Zscaler Zero Trust Cyber Associate (ZTCA)
- Deep Learning Specialization — DeepLearning.AI (Coursera)

RECOGNITION

- GATE — Joint All India Rank 2
- Two-time Apple Best Solution Award recipient
- IBM Bravo Award recipient